FFF Printing Fundamentals

Part 1: Process

3D Printing is a Process

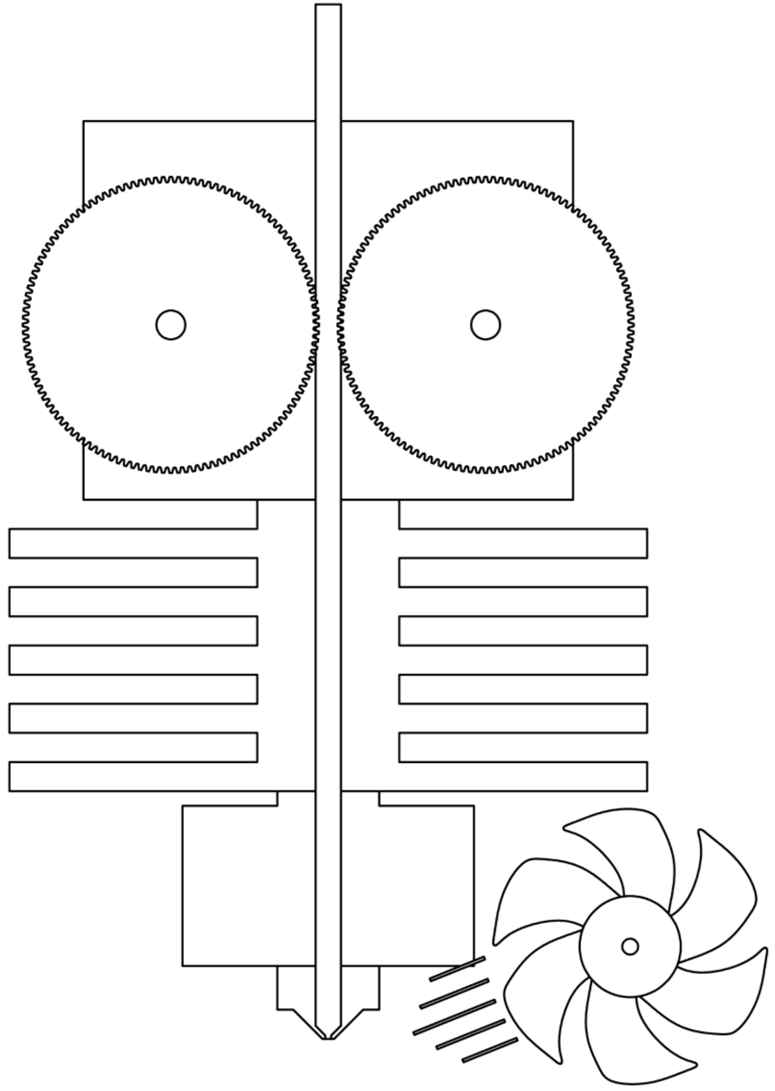
1) Start with a 3D design (digital file).
   - If you download, check the license for restrictions.
   - **STL**. Approximates shapes with tiny triangles.
   - **STEP**. Real geometry. Great for 3D modeling.
   - **3MF**. Compressed container, like a zip file.
     Multiple objects, slicer settings, multi-color options, …
2) Slice the design.
   - Printers do not accept 3D models.
   - Use software to “slice” objects into vertically-stacked layers.
   - Translate shapes into to Gcode.

3D Printing is a Process

3) Print.
    - Do you need to modify the design?
    - Or tune printer settings?
4) Post-process.
    - Remove supports.
    - Assemble.
    - Glue.
    - Filler + sanding to remove layer lines.
    - Paint.

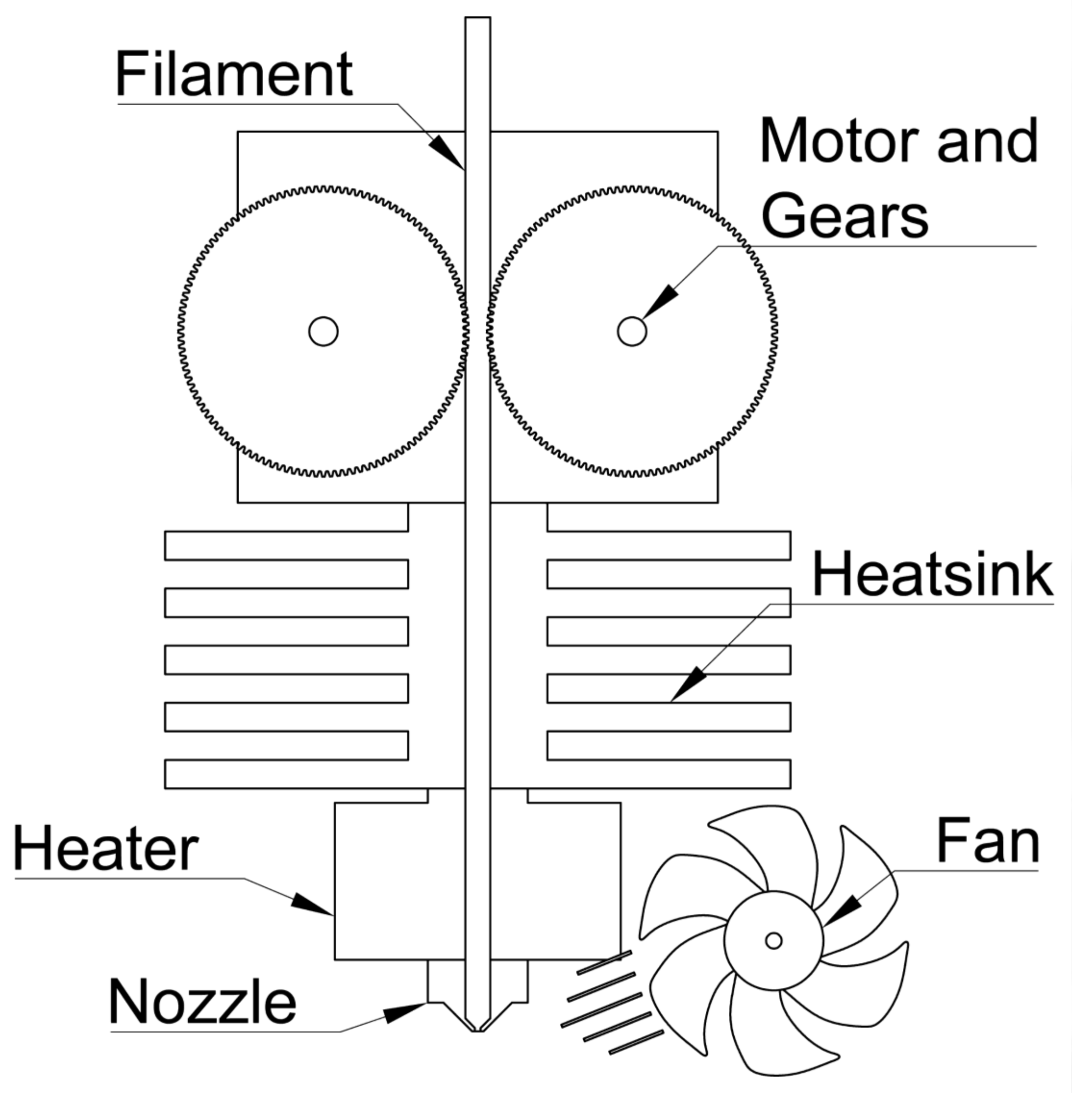
Pushing Plastic

- What kind of 3D printing?
- Filament and extrusion-based additive manufacturing.
- **FFF. Fused filament fabrication.**
- FDM™. Fundamentally the same but owned by Stratasys.
  - Fused deposition modeling.
- Toolhead. Moves around and deposits plastic.



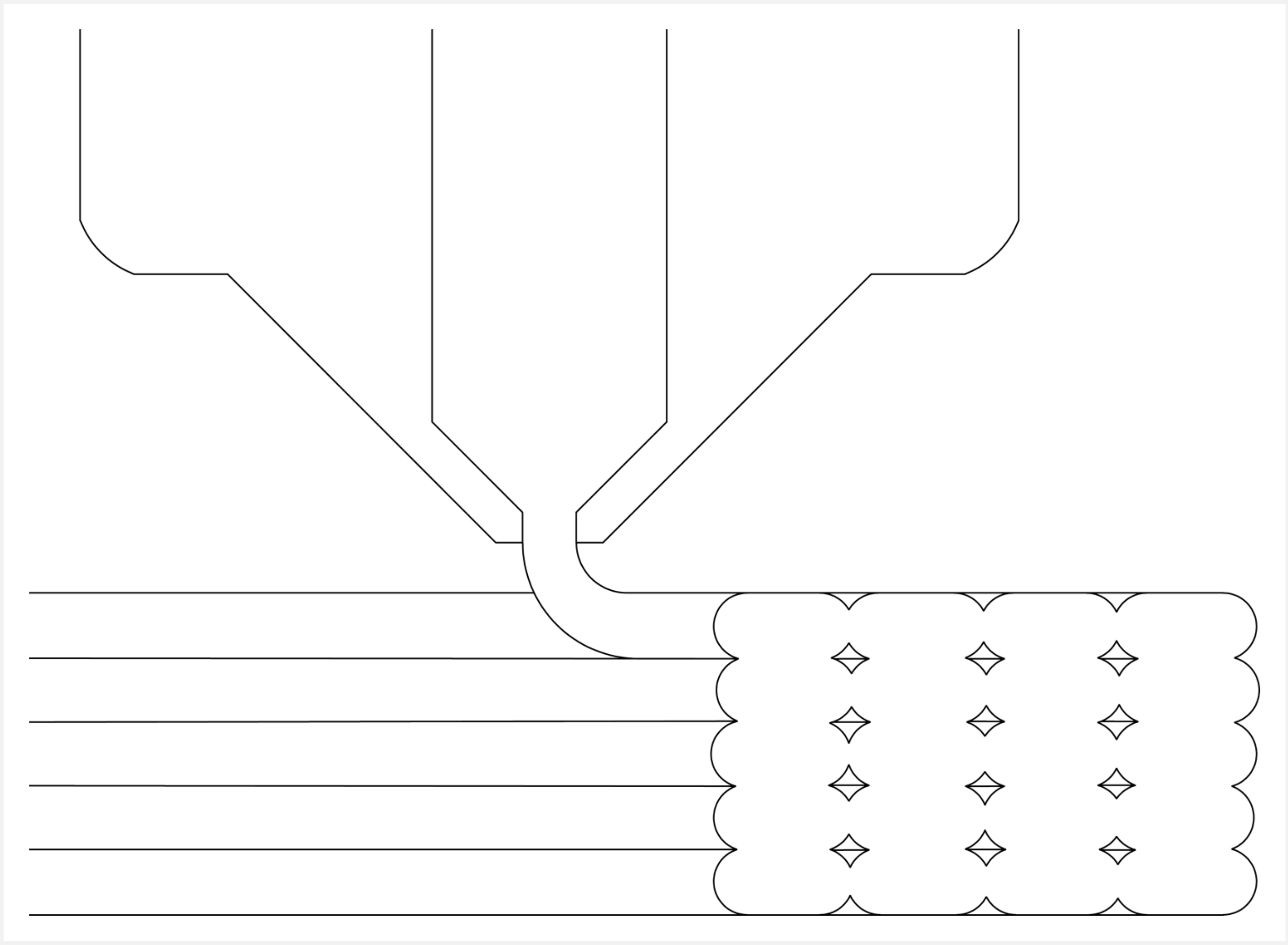
Pushing Plastic

- Filament. Long, continuous plastic rod.
- Motor and gears grab and push filament.
- Heatsink keeps heat in the hotend (heater + nozzle).
- Heater softens filament at the glass transition temperature.
- Nozzle opening shapes plastic.
- Fan cools hot plastic.



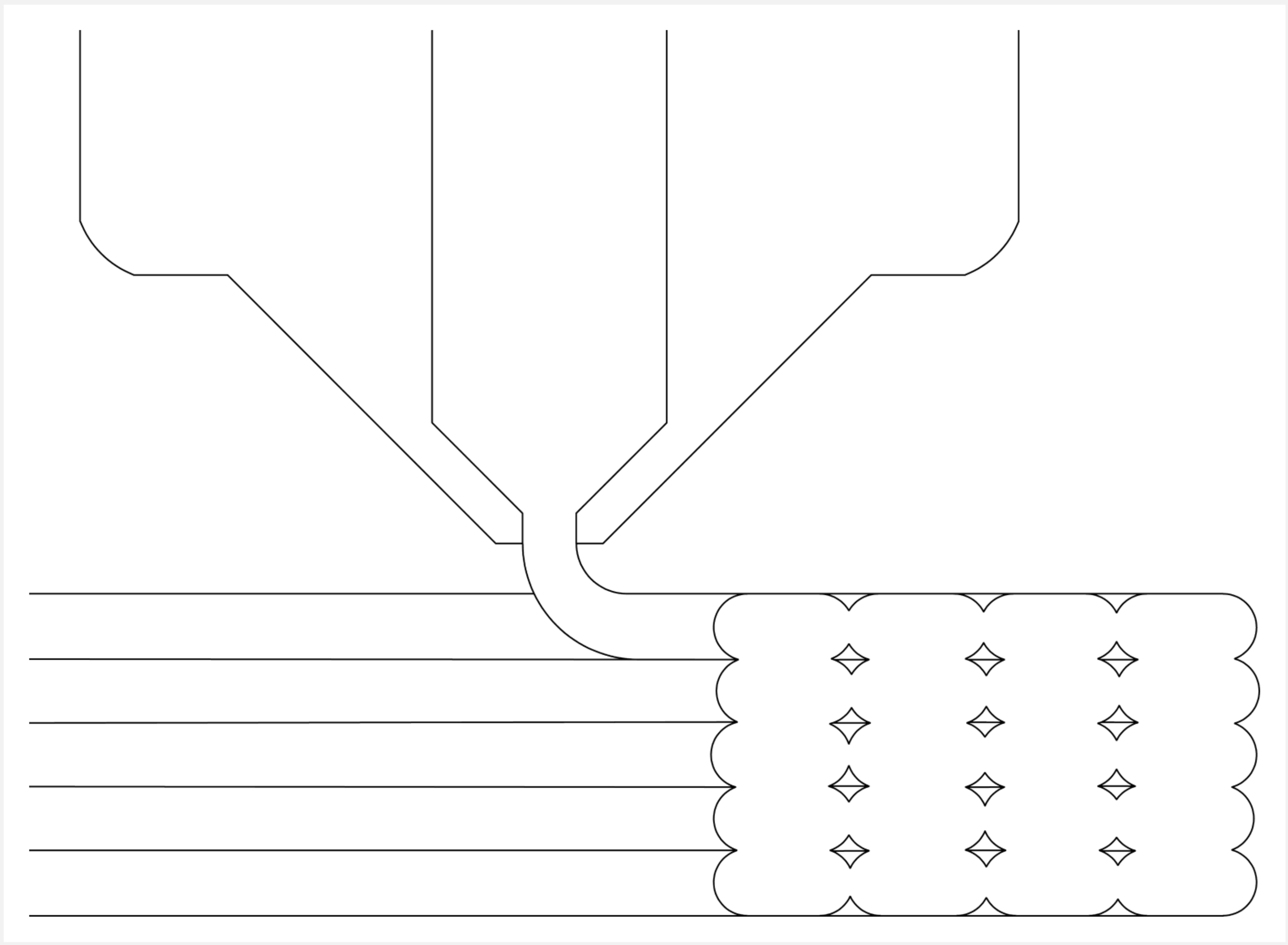
Extrusions

- Toolhead deposits lines called **extrusions**.
- Best to squish an extrusion into place.
- Lock in place as they cool.



Stacking Layers

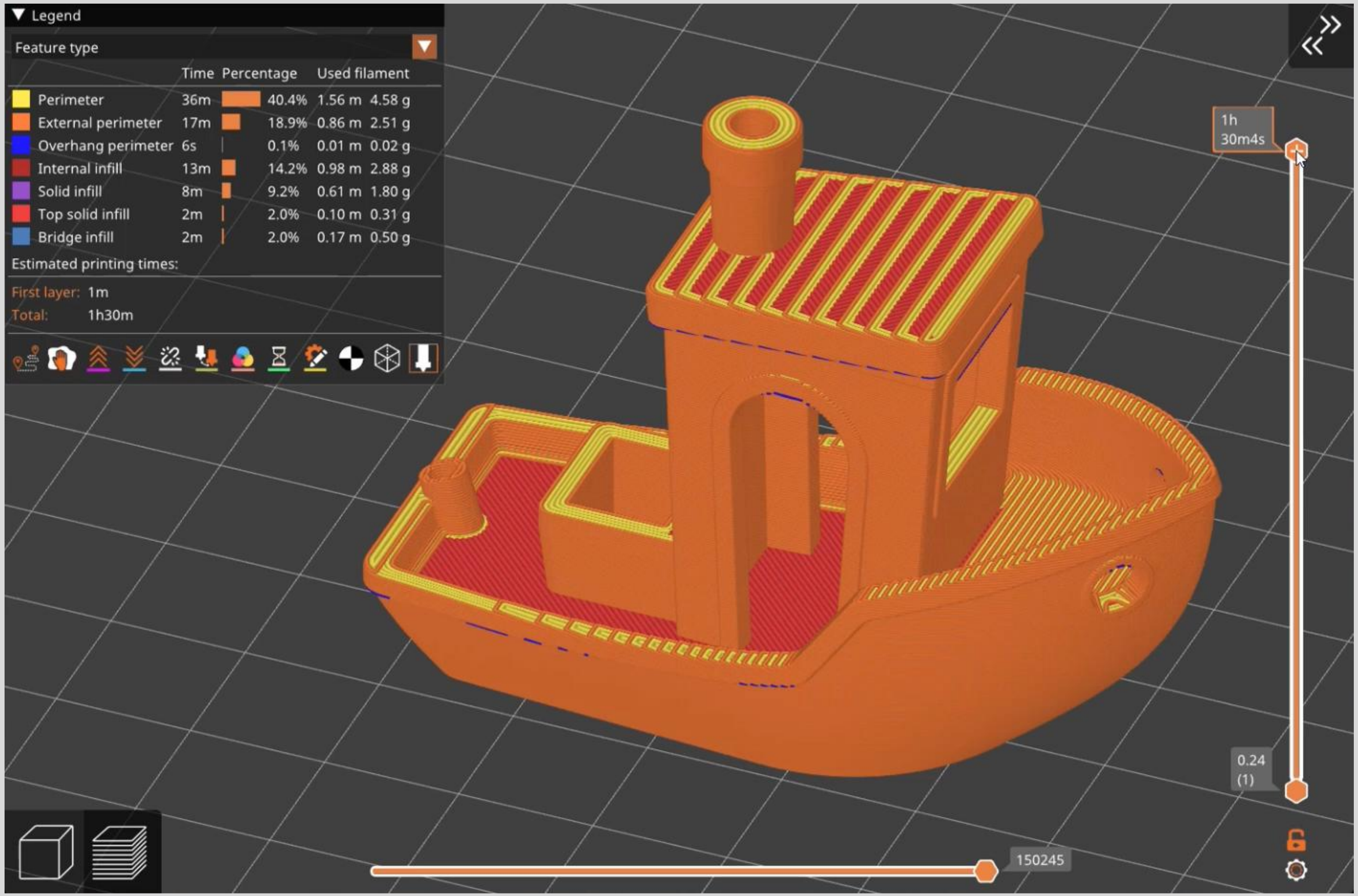
- Printers **simulate** 3D shapes by stacking layers.
- You could call it 2.5D printing.



Slicer Layers

Let's see how PrusaSlicer handles layers.

- This boat starts as a solid object.



(Animation, FlyingGyroscope GitHub)

Slicer Layers

- Slicers sort extrusions by shape and geometry.
- Color-coded.
- Adjust settings for each category.

Legend

Feature type

	Time	Percentage	Used filament
Perimeter	36m	40.4%	1.56 m 4.58 g
External perimeter	17m	18.9%	0.86 m 2.51 g
Overhang perimeter	6s	0.1%	0.01 m 0.02 g
Internal infill	13m	14.2%	0.98 m 2.88 g
Solid infill	8m	9.2%	0.61 m 1.80 g
Top solid infill	2m	2.0%	0.10 m 0.31 g
Bridge infill	2m	2.0%	0.17 m 0.50 g

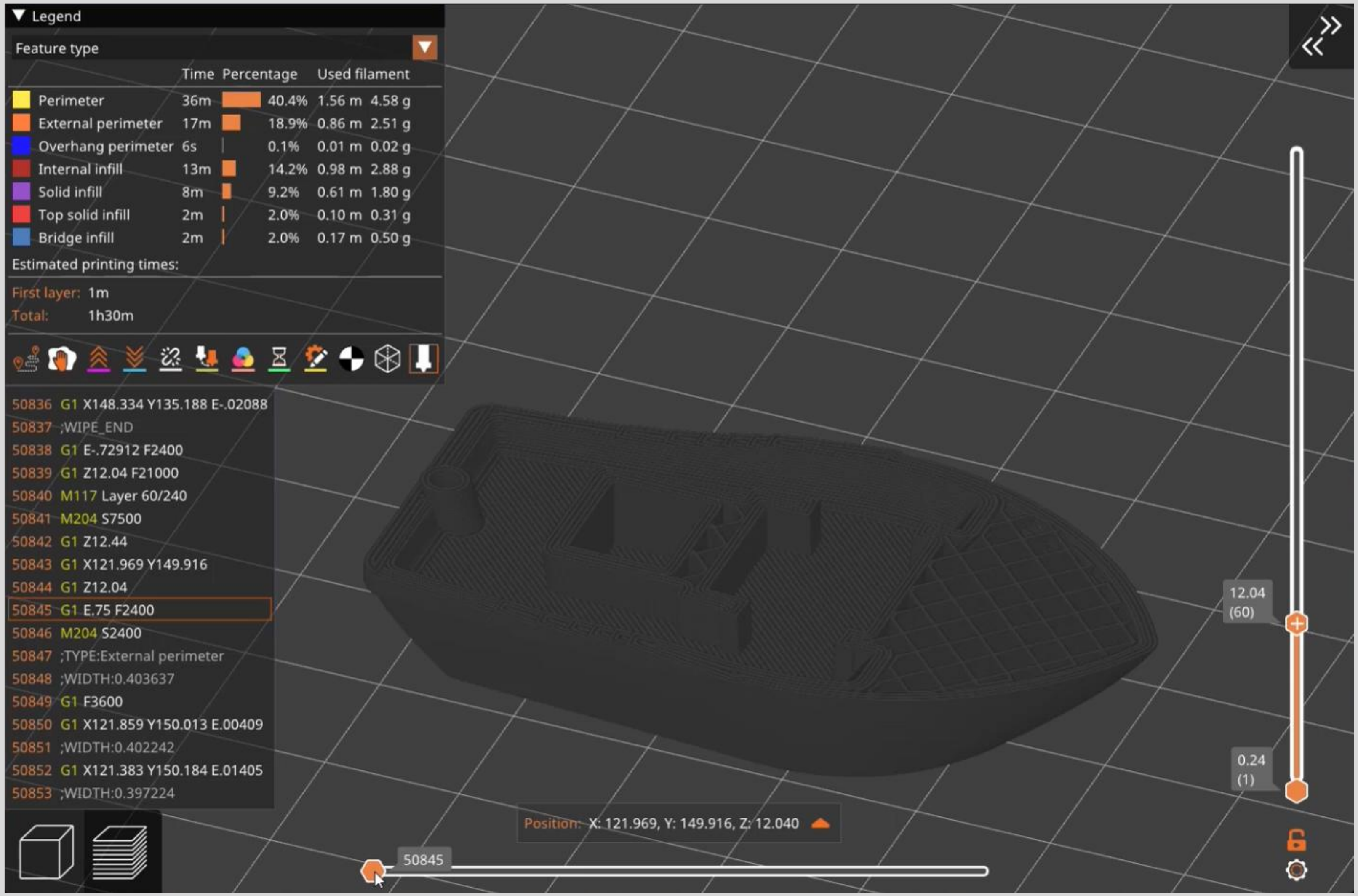
Estimated printing times:

First layer: 1m

Total: 1h30m

Slicer Layers

- Single layer preview.
- Gcode shows what the printer is actually doing (lower left).
- *G1* = move. Can include X, Y, or Z coordinates.
- *G1 X Y E* = move and extrude.



(Animation, [FlyingGyroscope GitHub])

(Animation, Thermal camera video by NeedItMakeIt)

Easy Filaments

- **PLA** (derived from corn starch).
  - General-purpose filament for indoor applications.
  - Excellent bridging.
  - Very stiff.
  - Deforms permanently under load (creep).
  - Low temperature resistance — will not survive inside a car on a hot summer day.
- **PETG** (plastic water bottles).
  - Pretty good general-purpose filament.
  - Hygroscopic (needs low-humidity storage).
  - Excellent choice for springs (elastic).
  - Moderate temperature resistance — can be used outdoors.

Easy Filaments

- What makes PLA and PETG easy?
  - Low warp (low shrink).
  - High dimensional accuracy.
  - Room temperature print chamber.

Proper Protocol

Let's review some everyday guidelines.

- Secure the loose end of a spool of filament at all times.
  - Tangles ruin prints and spools, and waste time.
- Do not reach into the printer while it is running.
  - Use the touchscreen or buttons to pause and resume prints.
- Watch the first layer before you walk away.
- Avoid burns.
  - Read nozzle and print bed temperatures on the display.
  - Never reach near a hot nozzle (210 °C = 410 °F).
- Do not leave fingerprints on print sheets.